

Dylan Pierre

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PROFESSIONAL SUMMARY

Computer Science graduate with a background in systems and backend software development. Experienced in designing microservices, systems programming, and building scalable backend services. Focused on writing clean, efficient code that solves real problems and drives measurable impact.

SOFTWARE ENGINEERING EXPERIENCE

Redis-mini | *Go, Concurrency, Backend, Databases*

- Built a concurrent, in-memory key-value database in Go, using goroutines and read/write mutexes to serialize writes while allowing concurrent reads, fed by a hand-written RESP protocol parser.
- Designed unit tests covering improper input and state corruption.

C-Like Language Interpreter | *C++, Systems*

- Designed and built a full language interpreter in C++ from scratch, taking source code from raw text to running output.
- Implemented every stage of the language pipeline: a state-driven tokenizer for lexical analysis, a recursive descent parser, an abstract syntax tree to represent program structure, a symbol table for scope resolution, and the tree-walking evaluator.

Secure API Gateway Microservice | *Go, Microservices, API, Security*

- Designed and implemented a secure authentication/authorization gateway microservice in Go as a RESTful API.
- Implemented PASETO tokens and Argon2 password hashing to protect user credentials and prevent unauthorized access.
- Developed role-based access control middleware that authorizes every request, scoping data and actions to each user's permission level.

Cervical Spine Fracture Detection | *Python, ML*

- Built a two-stage deep learning pipeline for cervical spine fracture detection, combining YOLO-based vertebrae localization with a TensorFlow classification model for per-vertebra fracture identification.
- Collaborated in a team with a fast-paced timeline, held weekly meetings, delegated responsibilities across project objectives, and delivered a working end-to-end pipeline.

RELATED EXPERIENCE

DataAnnotation

May 2024 – Present

Code Quality Analyst

Remote

- Reviewed and debugged AI-generated code across Python, JavaScript, and C++, identifying logic errors, security flaws, and inefficiencies against detailed technical specifications.
- Improved LLM capabilities through RLHF, documenting improvement areas and implementation strategies.

EDUCATION

Sonoma State University

3.5 GPA

Bachelor of Science in Computer Science

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Programming Languages, Software Engineering, Machine Learning

CERTIFICATIONS

CompTIA Security+ (Earned April 2026), **Google Cybersecurity Professional**

SKILLS

Programming Languages: C++, C#, Go, Python, Java, JavaScript, TypeScript, PowerShell

Tools & Methodologies: Git/Github, Agile/Scrum, RESTful APIs, Object-Oriented Programming, CI/CD, VSCode

Cloud & Infrastructure: AWS, Docker, SQL

Operating Systems: Windows, Linux, macOS